Radiation-Matter Interaction – Master M1 NE
TP GEANT4

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Installation of GEANT4 application

The GEANT4 documentation is available on the CERN web https://geant4.web.cern.ch/. GEANT4 is already installed on a common disk readable for all users having a computer account at IMT Atlantique.

- In this TP, you will create your own application by copying the GEANT4 example from the moodle in your *geant4* directory:

  Create your *geant4* directory: *mkdir geant4*
  Then: *cd geant4*
  Download the GEANT4 code from
  https://moodle.imt-atlantique.fr/course/view.php?id=2554
  decompress the file and put the obtained *B2a* directory in your *geant4* directory (as you did for the C++ exercise)

- To define the GEANT4 environment in your *geant4* directory write SETUP GEANT4:
  You will have to type this command every time you open a new shell terminal.

- To build your application, go to your *B2a* directory and then create a new directory: *mkdir B2a-build*

- Go to this new directory: *cd B2a-build*
  This new directory will be used to build your application i.e. to compile your program and create an executable file.

- Generate the Makefiles needed to build the *B2a* application:
  cmake ../

- To compile your application and create the executable file, you should type in the *B2a-build* directory: *make*

- You can now run your application in the *B2a-build* directory: *./exampleB2a*

Whenever you need to modify the source code, you have to recompile by typing *make* in the *B2a-build* working directory.

GEANT4 Simulation

Take time to carefully analyze the output text of the execution of your application, it will help you to understand how the "tracking" is working in GEANT4.

- When you launch *exampleB2a*, the macro *vis.mac* is automatically called by your GEANT4 application. Open the file which is in the current *B2a-build* working directory by typing: *kate vis.mac &*
 and try to understand the list of command lines written in this file.

- You can run all these command lines directly in interactive session. Try, for example to type: */vis/viewer/set/viewpointThetaPhi 50 120 deg* after the prompt *Idle*. The viewing angle of the device must be changed.

- To visualize the histograms showing the results; in your *xterm* launch
root simulation.root, then type *new TBrowser*, then click on *ROOT Files,* then click on histogram *fH1*. The ROOT software documentation is available at the following address *http://root.cern.ch/*.

The objective of this TP is to understand the GEANT4 software as Monte Carlo simulation of the radiation-matter interaction processes. For this, an experimental device consisting of a lead target *(Target)* and a liquid xenon detector *(tracker)* inside a volume of air *(World or Mother volume)* has already been implemented. To know the characteristics of this device, look at the classes defined in the *src* directory. You will have to modify the source code to define the simulation configuration and parameters (geometry, filling of histograms ...)

Simulation of 1 MeV γ-rays

1. Plot the histogram of energy deposition in liquid xenon detector without lead. To do this, type e.g. */run/beamOn 100000* at the command prompt *Idle>* to generate 100 000 gamma of 1 MeV. Repeat the simulation for different thicknesses of the volume of liquid xenon (1 cm, 10 cm, ...). The thickness of xenon is already defined in the class *B2aDetectorConstruction.cc* at the following code line:

```
trackerLength = 20*cm;
```

   What do you observe?

Important remark: to accelerate the simulation of a large number of events ~10^5, it is better to remove the visualization interface and the text output. To do this, you can create your own *run.mac* file with only the following parameters inside:

```
/run/initialize

# Define level of verbosity
/tracking/verbose 0

# Define particle type
/gun/particle gamma
# Define particle energy
/gun/energy 1 MeV
# start run with one particle generated
/run/beamOn 100000
```

To use this *run.mac* file to initialize the simulation parameters instead of the default *vis.mac* file you should launch: *./exampleB2a run.mac*

2. Add the lead target already defined in *B2aDetectorConstruction.cc*, to do this uncomment the following lines of code:

```
// new G4PVPlacement(0,                 // no rotation
//                  positionTarget,  // at (x,y,z)
//                  fLogicTarget,    // its logical volume
//                  "Target",        // its name
//                  worldLV,         // its mother volume
//                  false,           // no boolean operations
//                  0,               // copy number
//                  fCheckOverlaps); // checking overlaps
```

Plot the histogram of energy deposition in liquid xenon. Is the effect of the target in agreement with what you can predict from radiation-matter calculation?

<u>Simulation of 2 MeV electrons</u>

Simulate 2 MeV electrons with and without lead target. Plot the histogram of energy deposition in liquid xenon in both cases. Comment on the effect of the lead target on the number of detected electrons and the shape of the energy spectrum.